Apple Inc. (AAPL) Historical Stock Market Analysis

Week 6 Internship Project: Advanced Python Analysis

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Tools: Python, Pandas, NumPy, Matplotlib, yfinance

1. The Business Problem

Can five years of Apple stock data show how the price trends, when trading activity spikes, how risky the stock is, and whether any months behave differently? Investors and analysts need clear, data-backed answers to decide when to hold, trim or add exposure.

2. Dataset and Method

Daily AAPL prices were downloaded from Yahoo Finance with the yfinance library: 5 years, 6 August 2021 to 4 August 2026, covering Open, High, Low, Close, Adjusted Close and Volume. The analysis used Pandas for cleaning and feature engineering and Matplotlib for visualization.

3. Data Cleaning

The raw file loaded as 1,254 rows and 8 columns with every price column stored as text. These steps were applied:

- **Removed the stray header row.** The yfinance download added an extra row containing the ticker 'AAPL' in every column (and a missing date), which was dropped.
- **Dropped the redundant index column** created when the file was saved and reloaded.
- **Converted Close, High, Low, Open and Adj Close to numeric** and **Volume to integer**, because they had loaded as text (object).
- **Converted Date to datetime** so time-based calculations and charts work correctly.
- **Checked for missing values:** none remained after cleaning.
- **Sorted chronologically** so rolling averages and volatility are calculated in the right order.
- **Result:** a clean dataset of 1,253 trading days and 7 columns.

4. Feature Engineering

- **Daily Price Change** = Close minus Open (USD).
- **Percentage Price Change** = (Close minus Open) / Open x 100.
- **Year, Month and Month Number** extracted from Date for seasonal analysis.
- **7-day and 30-day moving averages** of Close, to separate short-term moves from the longer trend.
- **30-day volatility** = rolling standard deviation of Percentage Price Change.

5. Key Statistics

Metric	Value
Trading days analysed	1,253
Average closing price	$199.12 (median $187.87)
Highest close	$340.08 on 28 Jul 2026
Lowest close	$125.02 on 5 Jan 2023
Highest volume	318.68M shares on 20 Sep 2024
Average daily volume	64.7M shares (median 56.4M)
Average daily change (open to close)	0.12% (std. dev. 1.50%)
Largest daily move	-7.65% to +15.64%
30-day volatility	Average 1.40%, range 0.64% to 3.87%

6. Insights

1. **The price more than doubled.** Apple rose from a first close of $146.14 (Aug 2021) to a peak of $340.08 (28 Jul 2026), about +133%, and the close sat above both the 7-day ($330.27) and 30-day ($310.46) averages at the peak, which shows a sustained uptrend rather than a one-day spike.
2. **The 2023 low was followed by a very strong recovery.** The close bottomed at $125.02 on 5 Jan 2023, well below its 30-day average of $138.49 during the broad 2022 market sell-off, and the stock then climbed about 172% to its peak.
3. **The heaviest trading day was not a big price day.** Volume hit 318.68M on 20 Sep 2024, over five times the median, yet price moved only -0.77%, which points to a mechanical event such as index rebalancing or options expiry rather than company news.
4. **Risk arrives in bursts.** 30-day volatility averaged 1.40% but swung between 0.64% and 3.87%, with 75% of readings below 1.70%, because market-wide shocks hit in short clusters while calm periods dominate. The largest single-day gain (+15.64%) is consistent with the April 2025 tariff-pause rally, though the date should be confirmed in the data.
5. **July, November and October were the strongest months.** Average daily gains were +0.32%, +0.28% and +0.25%, while September was the weakest at -0.14%, but with only five years of data this pattern is suggestive, not proven.

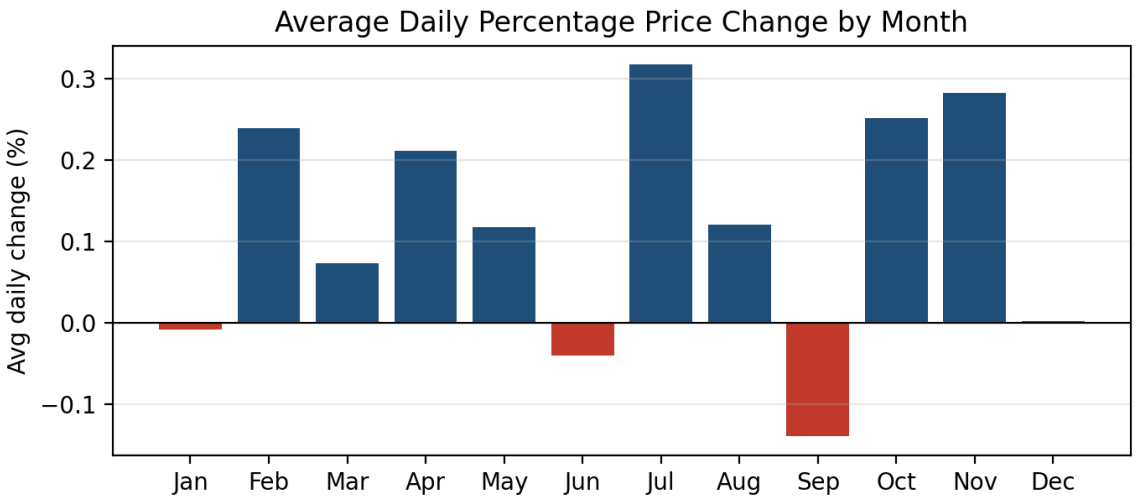


Figure 1: Average open-to-close daily change by month (values from notebook section 8.1).

7. Recommendations

1. **Use moving averages as a trend filter.** Treat price holding above the 30-day average as confirmation of an uptrend, and a close below it as a prompt to review the position.
2. **Read volume together with price.** Do not treat a volume spike as a signal unless price moves with it; flat-price spikes such as 20 Sep 2024 are usually mechanical.
3. **Size positions by volatility.** Consider reducing exposure when 30-day volatility climbs above about 1.70% (its 75th percentile) and allow larger positions in calm periods.
4. **Do not trade on seasonality alone.** Validate the monthly pattern on 10+ years of data and on close-to-close returns before relying on it.
5. **Extend the analysis.** Compare AAPL with the S&P 500, add indicators such as RSI and Bollinger Bands, and backtest any rule before using it.

8. Conclusion

Python, Pandas and Matplotlib turned raw, messy Yahoo Finance data into clear findings on trend, activity, risk and seasonality. This report is for learning purposes and is not financial advice.